

Analyzing the Unmasking Effect in China: Linking Temperature, Air Pollution and Sunspot Activity using Python



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Abstract:

In recent years China has accelerated a drastic cut in emissions, commonly referred to as “unmasking”, which has provided a unique opportunity to study the effects of anthropogenic emission on local climate. This study aims to enhance the understanding of unmasking by linking local temperature, air pollution and global sunspot activity. Data was taken at the start of each month with air quality data from the World Air Quality Index of PM2.5, PM10, O2, NO2, CO, and SO2 . Temperature data was taken from the ERA5 dataset at 2 meters above surface level. Sunspot data was collected through global daily sunspot numbers from NOAA. Monthly Pearson correlation coefficients were calculated to assess relationships between sunspots and temperature, as well as temperature and air quality. In order to effectively analyze the entire country of China 7 cities were chosen with diverse temperature ratings, air quality indexes, and geographical relationships which were Beijing, Changsha, Hangzhou, Lhasa, Shahepu, Shanghai, and Xian.

Our results highlight regional and particulate matter differences, with a strong inverse correlation between air quality and temperature. Specifically, out of the 34 pearson correlation coefficients run across air quality data (PM2.5, PM10, NO2, CO, and SO2) and temperature between various Chinese cities, 95% displayed a negative relationship. In contrast, the Pearson correlation coefficient between sunspot activity and temperature across all 7 cities was positive and low demonstrating a weak direct relation.

Therefore, temperature and sunspot data had little to no correlation suggesting that the recent rise in temperature is due to internal factors such as changes in air quality. This study is one of the first in the field to analyze this effect as it is only being initially observed and the effects are still ongoing. Overall, this study highlighted the effects that unmasking can have and suggests further research needs to be conducted on the cooling effect of certain air quality measures.

Background Information:

- Unmasking: refers to the reduction of anthropogenic emissions, particularly in highly industrialized regions, which “unmasks” underlying climatic trends that were previously obscured by pollution
- In recent years China has implemented stricter air quality laws such as *Air Pollution Prevention and Control Action Plan (2013)* and *Blue Sky Defense Action Plan (2018)*, leading to significant reductions in air pollution
- Sunspots are dark areas on the sun
 - It is debated whether they affect temperature in a positive relationship (Gray, L. J., et al. 2010)

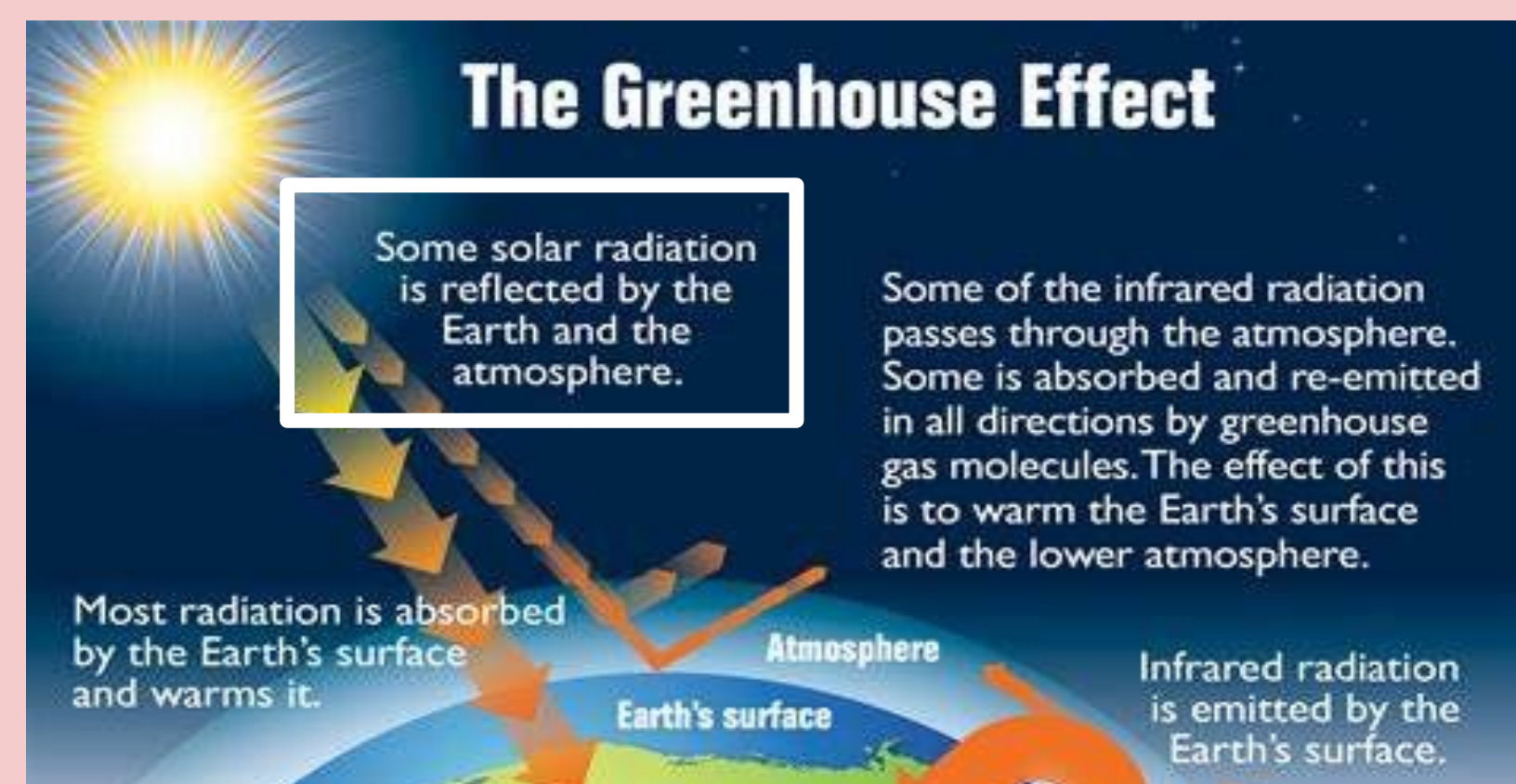


Figure 1: Typical Diagram of greenhouse gas effect (BYJO, 2024)

Method/Materials:

Chinese Cities

- 7 cities were chosen with diverse temperature ratings, air quality indexes, and geographical relationships
 - Beijing, Changsha, Hangzhou, Lhasa, Shahepu, Shanghai, and Xian.

NOAA Global Sunspot Data

- Global daily sunspot numbers were collected from the NOAA National Centers for Environmental Information (NCEI) to analyze solar sunspot activity

ERA5 Climate Data

- Temperature data was sourced from the ERA5 dataset,
 - provides climate data at 2 meters above surface level,
 - Focused on monthly averages for Chinese cities over the past 15 years

World Air Quality Index

- Air quality data was gathered from the World Air Quality Index,
 - PM2.5, PM10, O2, NO2, CO, and SO2,
 - data taken at the start of each month.

Statistical Analysis

- Data from NOAA and ERA5 was processed used Python
- Missing data points were filtered out using the data.table and NumPy libraries for data manipulation
- JupyterLab Notebook v 3.2.9 was used
- Pearson Correlation Coefficients from the past 10 years of the first of every month data set were used to assess relationship between
 - Sunspots and Temperature
 - Air Quality and Temperature

Results:

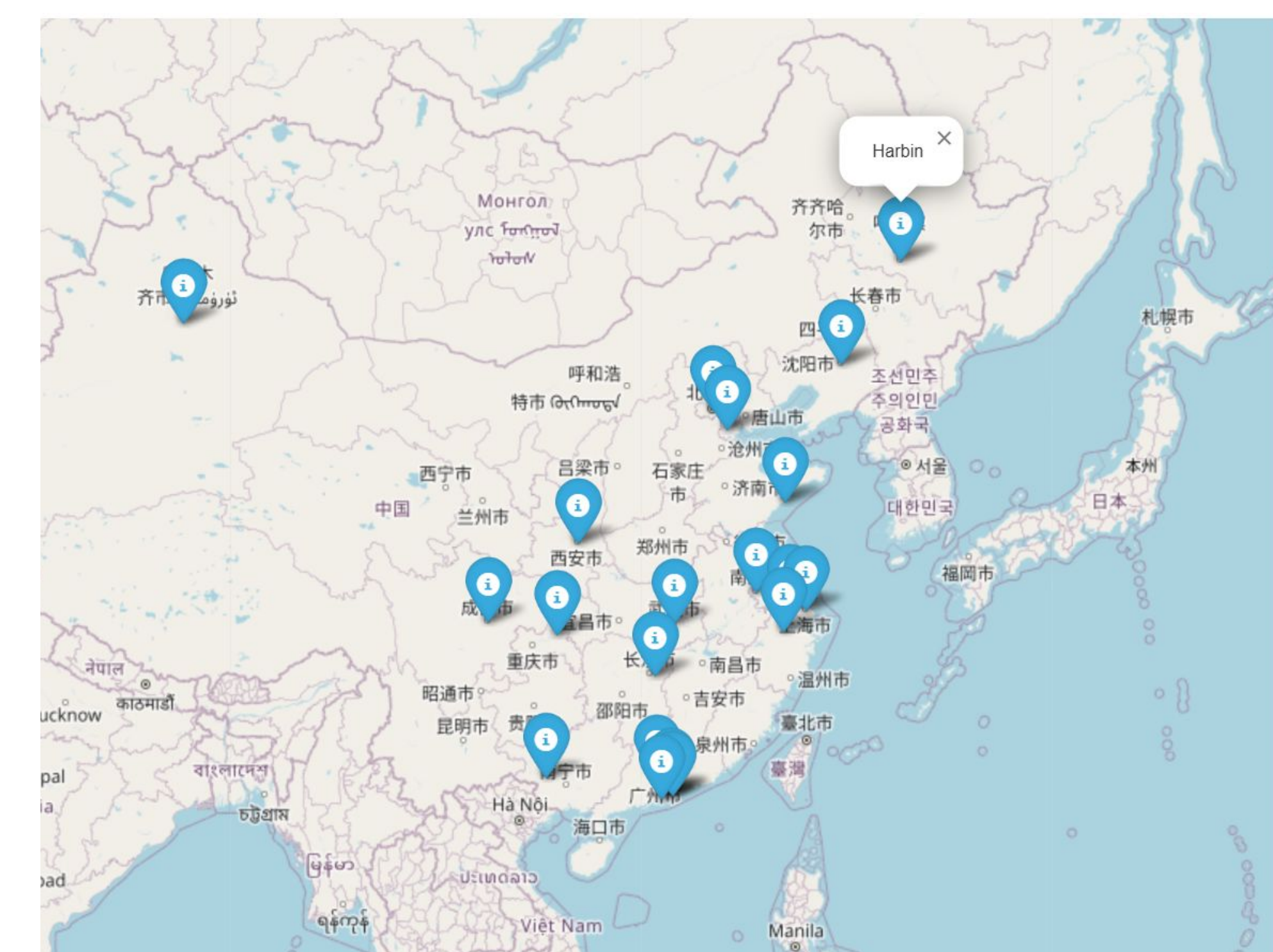


Figure 2: Map created using JupyterLab Notebook of potential cities for examination within China. Those cities include “Beijing”, “Shanghai”, “Guangzhou”, “Shenzhen”, “Chengdu”, “Xi’an”, “Hangzhou”, “Chongqing”, “Nanjing”, “Wuhan”, “Suzhou”, “Tianjin”, “Hong Kong”, “Macau”, “Qingdao”, “Harbin”, “Urumqi”, “Shenyang”, “Changsha”, and “Zhengzhou”. (from this study)

Pearson Correlation Coefficient between Air Quality and Temperature Across Various Chinese Cities

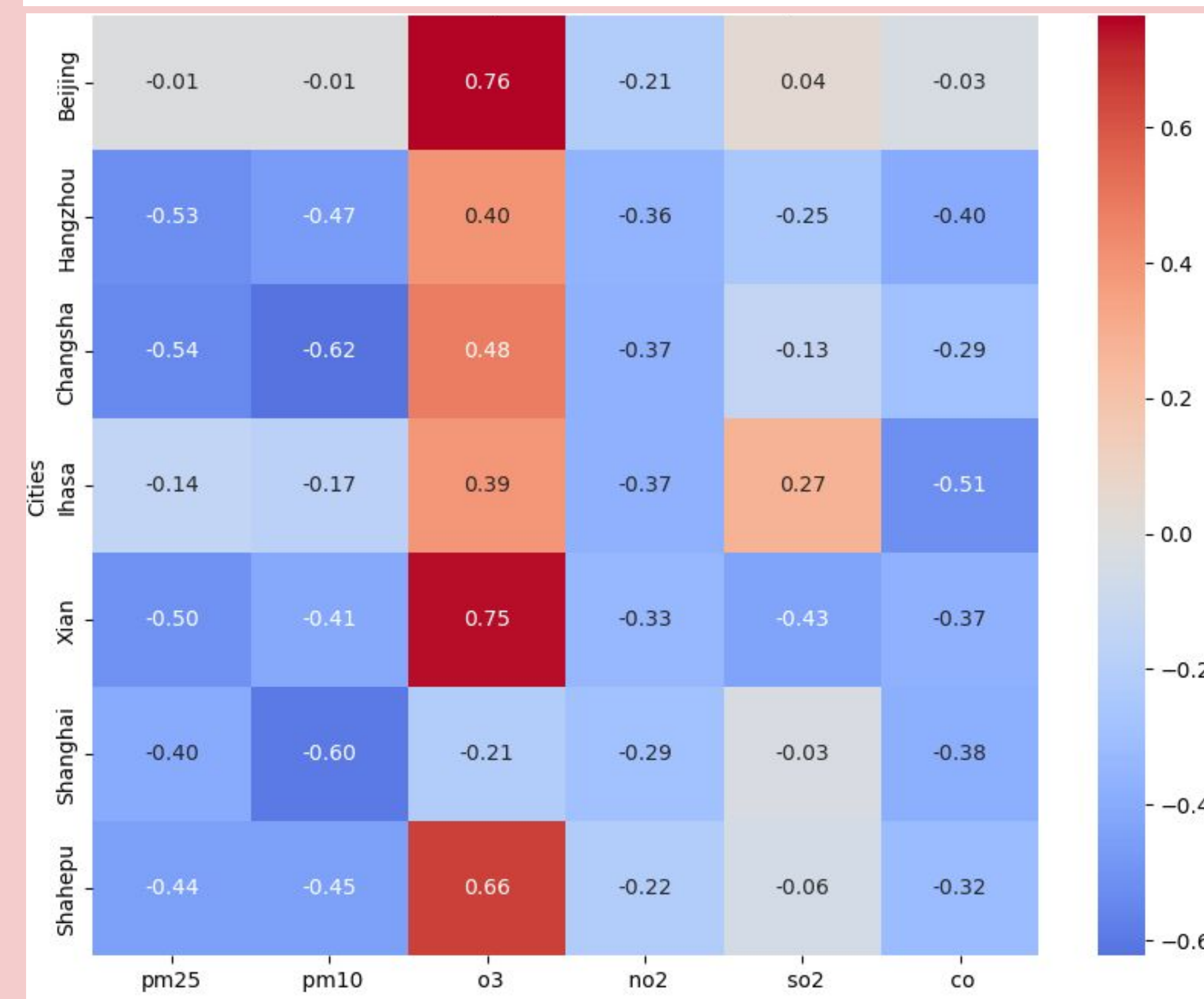


Figure 3: Heatmap of pearson correlation coefficient between air quality data and temperature across various cities. Blue indicates a negative correlation. Red indicates a positive correlation. (this study)

Pearson Correlation Coefficient between Air Quality and Sunspot Activity Across Various Chinese Cities

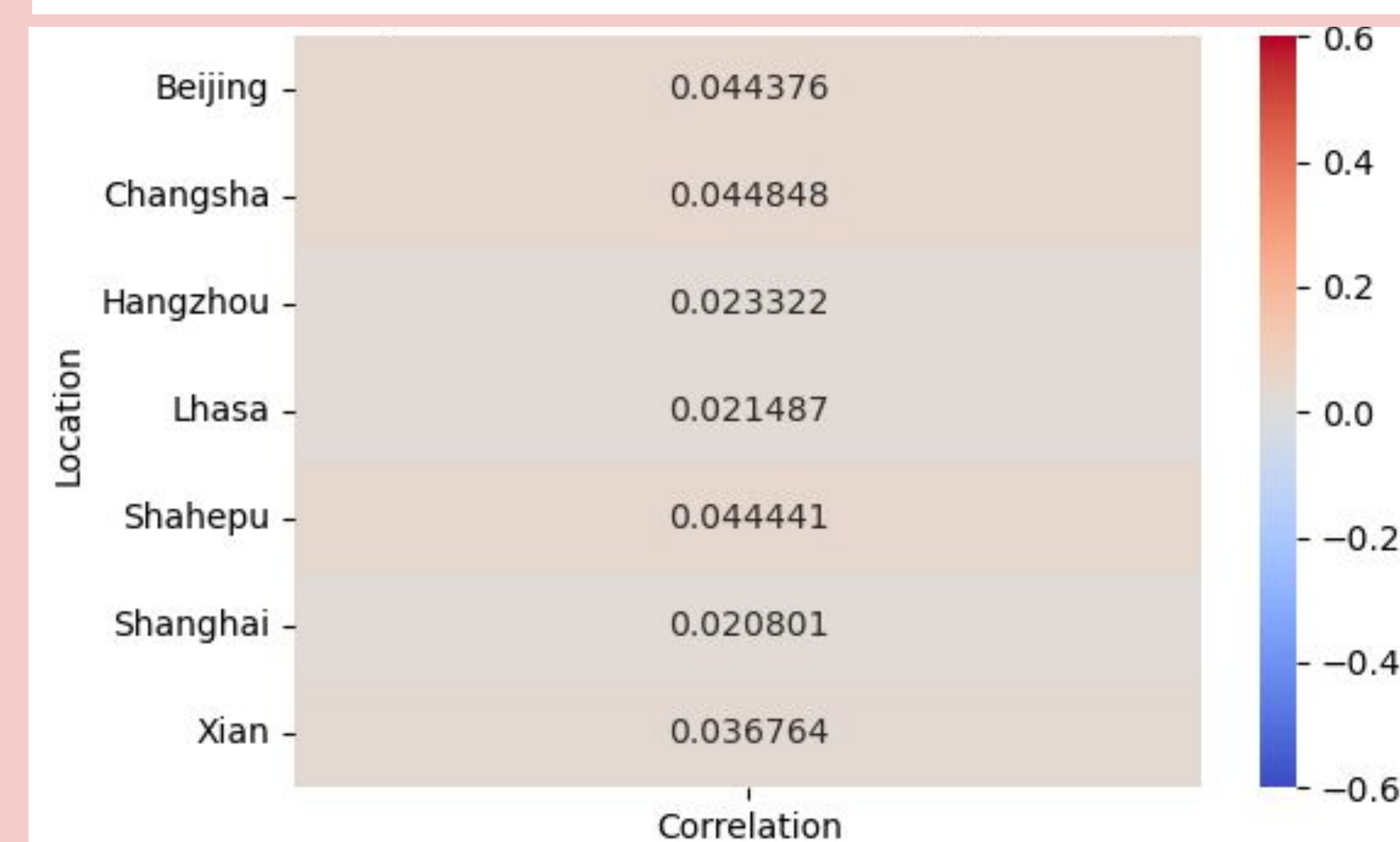


Figure 3: Heatmap of pearson correlation coefficient between sunspot data and temperature across various cities. Blue indicates a negative correlation. Red indicates a positive correlation. (from this study)

Discussion/Conclusion:

- Air Quality and Temperature are inversely related suggesting that particles like PM 2.5 and PM 10 might have a cooling effect
 - attributed to the different properties of aerosols which can scatter light and reduce surface temperature (Ramanathan, V., & Feng, Y. 2009)
- No significant correlation between sunspot activity and temperature activity
 - indicates that the changes in the increase in temperature are not due to external factors such as sunspot number
- Factors such as urbanization, industrial emissions and deforestation have a more direct effect on temperature decreases currently observed
- While current governance focuses on decreasing emissions this study shows a paradoxical relationship between
 - highlights the need for a new policy to be put in place to balance reduction and aerosols and temperature (Streets, D. G., et al, (2006).)
- There were certain limitations to this study
 - missing data points
 - the temporal resolution was .5 degrees
 - The study is limited to China so not applicable to the whole world
- Future Research:
 - More long term studies if the climate could bounce back or react differently
 - Urban heat islands could be a potential source for more specified look at increasing air pollution
- Overall, climate is a complex system and this study goes to show yet another layer of unique interactions.

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